Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Incorrect statement about $PH_3$is :

(A) It is produced by hydrolysis of Ca_3P_2.

(B) It gives black ppt. (Cu_3P_2) with $CuSO_4$ solution.

(C) Spontaneously burns in presence of P_2H_4

(D) It does not react with B_2H_6.

Q.2

The number of $P-OH$ bonds and the oxidation state of phosphorus atom in pyrophosphoric acid $(H_4P_2O_7)$ respectively are :

(A) four and four

(B) five and four

(C) five and five

(D) four and five

Q.3

P_2H_4 can be removed from phosphine containing traces of it

(A) by passing impure PH_3 gas through a freezing mixture.

(B) by passing the impure PH_3 gas through HI and then its treatment with KOH (aq).

(C) by both (a) and (b).

(D) by none of these.

Q.4

For H_3PO_3 and H_3PO_4, the correct choice is

(A) H_3PO_3 is dibasic and reducing

(B) H_3PO_4 is diabasic and non-reducing

(C) H_3PO_4 is tribasic and reducing

(D) H_3PO_4 is tribasic and non reducing

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Q.5

One mole of calcium phosphide on reaction with excess of water gives

(A) one mole of phosphine.

(B) two moles of phosphoric acid.

(C) two moles of phosphine.

(D) one mole of phosphorus oxide.

Q.6

Which of the following is a cyclic phosphate?

(A) $H_3P_3O_{10}$

(B) $H_6P_4O_{13}$

(C) $H_5P_5O_{15}$

(D) $H_7P_5O_{16}$

Q.7

P_4O_{10} is not used to dry NH_3 gas because

(A) P_4O_{10} reacts with moisture in NH_3.

(B) P_4O_{10} is not a drying agent.

(C) P_4O_{10} is acidic and NH_3 is basic.

(D) P_4O_{10} is basic and NH_3 is acidic.

Q.8

Which of the following is incorrect for white and red phosphorus ?

(A) They are both soluble in CS_2.

(B) They can be oxidised by heating in air.

(C) They consist of the same kind of atoms.

(D) They can be converted into one another.

Q.9

Which of the following oxy-acids has the maximum number of hydrogens directly attached to phosphorus?

(A) $H_4P_2O_7$

(B) H_3PO_2

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(C) H_3PO_3

(D) H_3PO_4

Q.10

Number of sigma bonds in P_4O_{10} is

(A) 6

(B) 7

(C) 17

(D) 16

Q.11

Which of the following compound has a P-P bond?

(A) $H_4P_2O_5$

(B) $(HPO_3)_3$

(C) $H_4P_2O_6$

(D) $H_4P_2O_7$

Q.12

$A + H_2O \longrightarrow B + HCl$
$B + H_2O \longrightarrow C + HCl$
Compound $(A), (B)$ and (C) will be respectively

(A) $PCl_5, POCl_3, H_3PO_3$

(B) $PCl_5, POCl_3, H_3PO_4$

(C) $SOCl_2, POCl_3, H_3PO_3$

(D) $PCl_3, POCl_3, H_3PO_4$

Q.13

Superphosphate of lime is obtained from the reaction of :

(A) Bones with gypsum.

(B) Calcium phosphate with sulphuric acid.

(C) Calcium phosphate with HCl.

(D) Calcium carbonate with phosphoric acid.

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Q.14

Phosphine is not obtained by which of the following reaction?

(A) White P is heated with $NaOH$.

(B) $Red\,P$ is heated with $NaOH$.

(C) Ca_3P_2 reacts with water.

(D) Phosphorus trioxide is boiled with water.

Q.15

Wheh of the following allotropic form of phosphorus is thermodynamcally most stable.

(A) White phosphorus

(B) Black phosphorus

(C) Red phosphorus

(D) All are equally stable

Q.16

Conc. HNO_3 is heated with P_2O_5 to form :

(A) NO_2

(B) NO

(C) N_2O_5

(D) N_2O

Q.17

The $P-P-P$ bond angle in white phosphorus is

(A) 120°

(B) 90°

(C) 60°

(D) $109^\circ 28'$

Q.18

Poisonous form of phosphorus is

(A) white

(B) red

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(C) black

(D) violet

Q.19

The number of $P-O-P$ bonds in cyclic metaphosphate ion is

(A) 0

(B) 1

(C) 3

(D) 4

Q.20

Which of the following metaphosphate ion is not known to exist in free state?

(A) PO_3^-

(B) $(PO_3)_2^{2-}$

(C) $(PO_3)_3^{3-}$

(D) $(PO_3)_4^{4-}$

Answers & Solutions

Q.1 Answer:

It does not react with B_2H_6.

Solution:

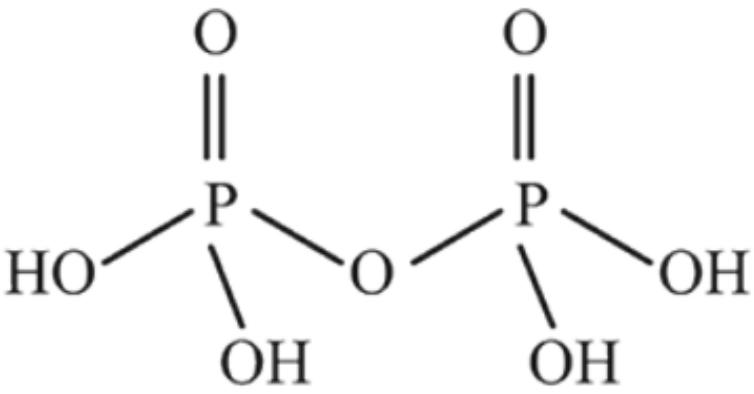
PH_3 **(Lewis base) can react with** B_2H_6 **(Lewis acid).**

Q.2 Answer:

four and five

Solution:

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Pyrophosphoric acid ($H_4P_2O_7$)

Oxidation State :

Each P **atom is bound to one oxygen** $= -1$ $2OH = -1 \times 2 = -2$

Oxygen $= -2$ **Total** $= -5$ $P = +5$

Q.3 Answer:

by both (a) and (b).

Solution:

Both the methods (a) and (b) can be used. When passed through freezing mixture the P_2H_4 **present condenses and pure** PH_3 **is obtained.**

When passed through HI, PH_3 **is absorbed forming** PH_4I.PH_4I **when treated with** KOH **(aq) yields pure phosphine.**

$$PH_3 + HI \longrightarrow PH_4I$$

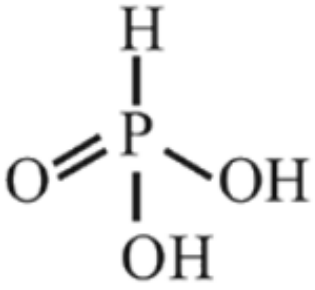
$$PH_4I + KOH(aq) \longrightarrow KI + H_2O + PH_3 \uparrow$$

Q.4 Answer:

H_3PO_3 is dibasic and reducing

Solution:

The structure of H_3PO_3 **is as given below.**



There are only two-OH groups and hence it is dibasic. The oxidation number of P **in this acid is** $+3$,

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whereas P may have $+5$ oxidation state also. Therefore, H_3PO_3 can be oxidised which means H_3PO_3 is a reducing agent.

Q.5 Answer:

two moles of phosphine.

Solution:

$$\underset{1\text{ mole}}{Ca_3P_2} + 6H_2O \rightarrow 3Ca(OH)_2 + \underset{2\text{ moles}}{2PH_3}$$

Q.6 Answer:

$H_5P_5O_{15}$

Solution:

$H_5P_5O_{15}$ or $(HPO_3)_5$. It is metaphosphoric acid which is a cyclic phosphate.

Q.7 Answer:

P_4O_{10} reacts with moisture in NH_3.

Solution:

$$\underset{\text{Acidic}}{P_4O_{10}} + 2H_2O \longrightarrow \underset{\text{meta phosphoric acid}}{4HPO_3}$$

$$\xrightarrow[\text{basic}]{NH_3} \underset{\text{meta ammonium phosphate}}{NH_4PO_3}$$

Q.8 Answer:

They are both soluble in CS_2.

Solution:

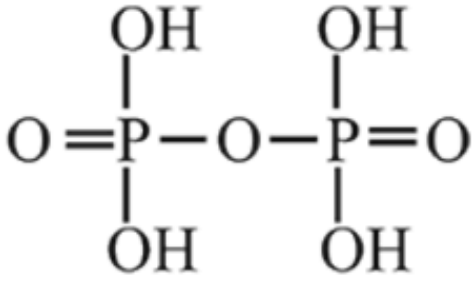
Both white and red phosphorus are not soluble in CS_2 only white phosphorus is soluble in CS_2

Q.9 Answer:

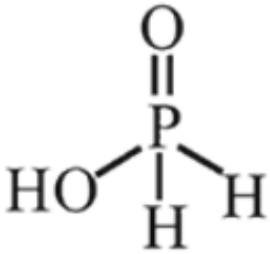
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H_3PO_2

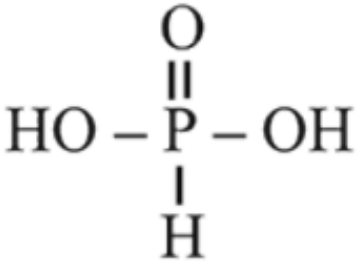
Solution:

(a) $H_4P_2O_7 \Rightarrow$ 

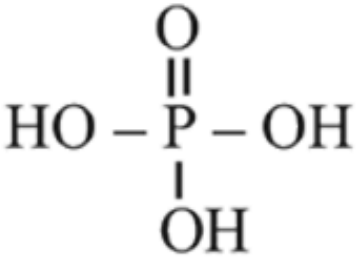
Pyrophosphoric acid

(b) $H_3PO_2 \Rightarrow$ 

Hypophosphorous acid

(c) $H_3PO_3 \Rightarrow$ 

Phosphorous acid

(d) $H_3PO_4 \Rightarrow$ 

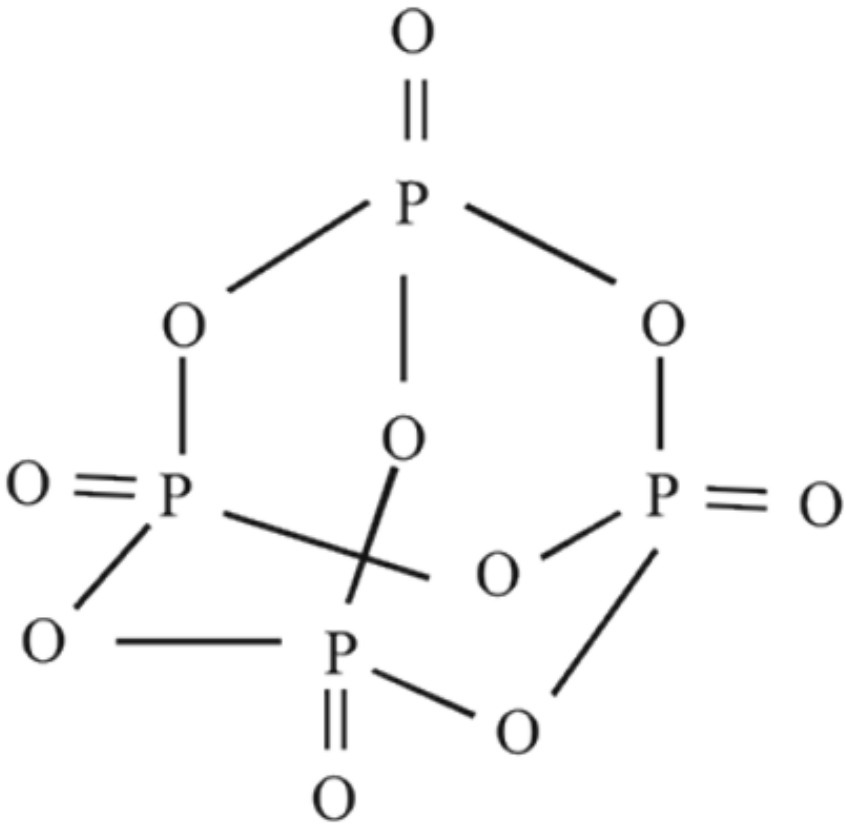
orthophosphoric acid

Q.10 Answer:

16

Solution:

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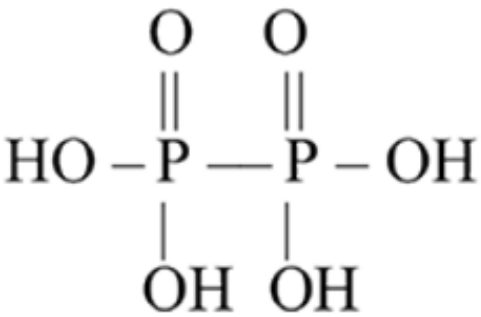


Q.11 Answer:

$H_4P_2O_6$

Solution:

$H_4P_2O_6$ has P–P linkage



Q.12 Answer:

$PCl_5, POCl_3, H_3PO_4$

Solution:

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$$\underset{(A)}{PCl_5} + H_2O \longrightarrow \underset{(B)}{POCl_3} + 2HCl$$

$$\underset{(B)}{POCl_3} + 3H_2O \longrightarrow \underset{(C)}{H_3PO_4} + 3HCl$$

Q.13 Answer:

Calcium phosphate with sulphuric acid.

Solution:

$$Ca_3(PO_4)_2 + 2H_2SO_4 + 5H_2O \rightarrow Ca(H_2PO_4)_2 \cdot H_2O + 2CaSO_4 \cdot 2H_2O$$

Q.14 Answer:

Red P is heated with NaOH.

Solution:

Red P does not react with $NaOH$ to give PH_3.

Q.15 Answer:

Black phosphorus

Solution:

Bleck Phosphorous is themodynamically most stable.

Q.16 Answer:

N_2O_5

Solution:

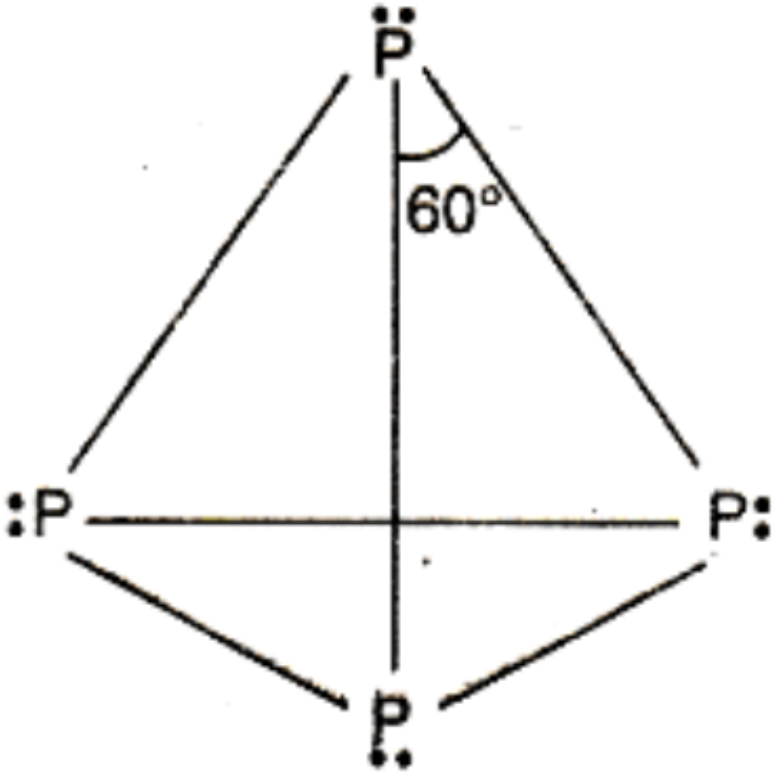
$$P_2O_5 + 2HNO_3 \rightarrow N_2O_5 + 2HPO_3$$

Q.17 Answer:

60°

Solution:

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Q.18 Answer:

white

Solution:

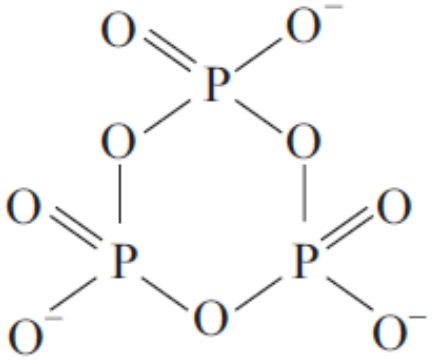
The white phosphorus allotrope is the most reactive, least stable and most toxic of the three. It consists of four phosphorus molecules that chemically bond to form a tetrahedral shape.

Q.19 Answer:

3

Solution:

The cyclic metaphosphate ion is



Q.20 Answer:

PO_3^-

Solution:

The free monometaphosphate ions does not exist. The metaphosphates form a family of ring compounds

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